NOVAMART

MARKETING ANALYTICS

END-TO-END DATA ANALYTICS PROJECT

A practical business case study using SQL and Power BI

Project prepared by	**Darshana Narjari**
Project start date	09-10-2026
Project submission date:	10-10-2026
Dataset:	NovaMart_Marketing_Analytics_100K.xlsx
Source Link:	NovaMart Marketing Analytics Dataset - GitHub Repository
Tools:	PostgreSQL • Power BI • DAX • Excel

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01 | Executive Summary

NovaMart is treated in this case study as a retail business running digital marketing campaigns across multiple channels, campaign types, regions, devices and audience segments. This analysis turns campaign-level data into evidence to understand where marketing investment generates revenue, where conversion performance is strongest, and where further investigation may be needed.

The project follows an end-to-end analytics workflow: inspect the source workbook, prepare and validate data, query the cleaned table in PostgreSQL, calculate performance metrics, and present findings in an interactive Power BI dashboard.

Core decision question: How can NovaMart use campaign performance evidence to improve marketing efficiency without relying on revenue totals alone?

Deliverables

- Documented data preparation and validation process.
- SQL analysis across channels, campaign types, campaign names, regions, devices, audiences and time periods.
- Power BI pages for overall performance, channel performance, campaign performance, and regional/audience analysis.
- Recommendations framed as actions to test and measure - not guaranteed outcomes.

Executive caution

The dataset shows unusually high ROAS values and large revenue totals relative to advertising spend. These figures should be validated before drawing business conclusions. ROAS measures revenue per unit of advertising spend; it is not profit.

02 | Business Context & Problem Statement

Business context

Digital marketing teams run campaigns across channels such as paid search, email, social media and influencer activity. Channels can generate different volumes of impressions, clicks, conversions, new customers, spend and attributed revenue. High revenue alone does not prove efficiency, and a high conversion rate alone does not prove the greatest business value.

Problem statement

NovaMart needs a consistent, evidence-based view of marketing performance to compare channels and campaigns, distinguish volume from efficiency, and identify areas for investigation. Before reallocating budget, the business also needs confidence that measures are defined and calculated consistently.

Why it matters

- A high-volume channel may have a high acquisition cost.
- A high click-through rate may still be followed by weak conversion.
- Regional or audience differences can be hidden by overall totals.
- Inconsistent units, duplicate rows or attribution issues can distort conclusions.

Scope

This is a descriptive and diagnostic project. It does not prove that a channel caused a revenue outcome or forecast future performance.

03 | Project Objectives & Business Questions

Objectives

1. Establish a reliable view of spend, impressions, clicks, conversions, revenue and new customers.
2. Compare channel performance using volume and efficiency metrics.
3. Identify differences by campaign type/name, audience, region, device and time period.
4. Surface potential underperformance and data-quality risks.
5. Communicate findings through an interactive Power BI report and action plan.

Business questions

- Which channels generate the most revenue and conversions?
- Which channels have the strongest CTR and conversion rate?
- Which channels show lower CAC and higher ROAS?
- How do campaign types and named campaigns compare?
- Do results vary by region, device or target audience?
- How does performance change by month and quarter?
- Are observed differences meaningful enough to justify a budget change?

04 | Dataset & Source Attribution

Working dataset

File: NovaMart_Marketing_Analytics_100K.xlsx

Cleaned SQL table: marketing_cleaned

Reported structure: approximately 100,000 records and 15 fields

Reported date range: 1 January 2024 – 31 December 2025

The dataset used in this project is available in the GitHub repository linked below. Its original source has not been independently verified.

Primary working-file repository: https://github.com/sachin-as-DA/marketing-analytics-powerbi-sql

Repository file: NovaMart_Marketing_Analytics_100K.xlsx.

05 | Data Dictionary

The table below describes the fields used in the analysis and their purpose.

Field	Meaning / analytical use
Campaign_Record_ID	Record identifier; check uniqueness and count records.
Campaign_Date	Date used for monthly and quarterly trends.
Campaign_Name	Named campaign for campaign-level comparison.
Channel	Marketing channel, e.g. Email, Google Ads, Instagram, Facebook, Influencer, YouTube.
Campaign_Type	Objective/type, e.g. Brand Awareness, Acquisition, Product Launch, Retargeting, Retention, Seasonal Sale.
Target_Audience	Audience segment targeted.
Region	Geographic grouping: Central, East, North, South, West.
Device_Type	Desktop, Mobile or Tablet.
Spend	Recorded marketing expenditure.
Impressions	Number of times campaign content/ad was displayed.
Clicks	Number of recorded clicks.
Conversions	Recorded conversion events; confirm business definition.
Revenue	Attributed revenue; units and attribution require validation.
New_Customers	Recorded newly acquired customers.
Campaign_Status	Status such as Active, Completed or Paused.

The exact meaning of one record should be confirmed from the source data before interpreting totals. This is important to avoid double-counting spend, revenue or conversions.

06 | Methodology & Workflow

Phase 1 - Inspect

1. Review workbook structure, row count, fields, data types, date coverage and summary statistics.
2. Identify nulls, duplicates, invalid dates, negative values and unexpected funnel relationships.

Phase 2 - Clean and validate

3. Standardise field names and data types where necessary.
4. Check record ID uniqueness and investigate duplicates before removing them.
5. Validate numeric fields and investigate missing, negative or implausible values.
6. Check funnel consistency: clicks generally should not exceed impressions; investigate exceptions.
7. Preserve a raw copy and document transformations.

Phase 3 - Analyse in PostgreSQL

SQL aggregations compared performance by channel, campaign type/name, region, device, audience, status, month and quarter. Both total results and performance rates were compared across marketing channels.

Phase 3 - Visualise in Power Bi

The cleaned PostgreSQL table was loaded into Power BI. DAX measures support ratio metrics. Separate pages cover overview, channel performance, campaign performance, and regional/audience analysis.

Phase 5 - Communicate

Translate findings into actions with a KPI and validation step. Test recommendations rather than assuming historical correlations prove causation.

07 | Data Quality, Cleaning & Validation

Validation checklist

- ☑ Record count and column count documented.
- ☑ Campaign_Record_ID uniqueness checked.
- ☑ Duplicates reviewed and documented.
- ☑ Nulls and blanks checked in dimensions and measures.
- ☑ Converted the campaign date into a proper date format and date range verified.
- ☑ Numeric fields checked for invalid values.
- ☑ Currency units and revenue attribution confirmed.
- ☑ Funnel relationships reviewed; exceptions investigated.
- ☑ Totals reconciled across workbook, PostgreSQL and Power BI.

The analysis showed unusually high ROAS values (around 497x) and revenue totals in the billions. These results require further validation. Revenue and Spend should use compatible units, duplicate revenue should be avoided, and ROAS should be calculated as total attributed revenue divided by total spend.

The same KPI definitions should be used in SQL and Power BI. Conversion Rate calculated by dividing total conversions by total clicks. Calculations should also handle cases where the denominator is zero.

08 | KPI Framework & Metric Definitions

KPI	Definition	Interpretation
CTR	Total Clicks ÷ Total Impressions	Share of impressions resulting in clicks.
Conversion Rate	Total Conversions ÷ Total Clicks	Share of clicks resulting in conversions.
CPC	Total Spend ÷ Total Clicks	Average recorded cost per click.
CPA	Total Spend ÷ Total Conversions	Average recorded cost per conversion.

CAC	Total Spend ÷ Total New Customers	Average recorded acquisition cost per new customer.
ROAS	Total Attributed Revenue ÷ Total Spend	Revenue per unit of spend; not profit.
Revenue	Sum of attributed revenue	Revenue volume; validate units and attribution.
New Customers	Sum of recorded new customers	Acquisition volume, subject to field definition.

Example DAX measures

```
CTR = DIVIDE(SUM(marketing_cleaned[Clicks]), SUM(marketing_cleaned[Impressions]), 0)
```

```
Conversion Rate = DIVIDE(SUM(marketing_cleaned[Conversions]),
SUM(marketing_cleaned[Clicks]), 0)
```

```
ROAS = DIVIDE(SUM(marketing_cleaned[Revenue]), SUM(marketing_cleaned[Spend]))
```

```
CAC = DIVIDE(SUM(marketing_cleaned[Spend]), SUM(marketing_cleaned[New_Customers]))
```

09 | Analysis Findings

These findings reflect the completed SQL exploration recorded for the project.

Channel performance

- Email showed the strongest CTR and conversion rate in the channel comparison and much higher reported ROAS with lower CAC than other channels.
- Google Ads generated the highest total revenue, making it important for revenue volume even though its efficiency profile differs from Email.
- YouTube showed comparatively weak CTR/conversion performance and high acquisition cost; investigate audience fit, creative, targeting and attribution before changing investment.

Campaign type and name

- Brand Awareness had the highest reported revenue among campaign types, while Retention showed the highest reported ROAS in the recorded comparison.
- Summer Sale generated the highest reported campaign revenue; Diwali Mega Sale had the highest reported ROAS among named campaigns.
- Some ROAS differences are small; ranking alone does not prove a meaningful or repeatable performance difference.

Audience, region, device and time

- West had the highest reported regional revenue; North showed the highest regional ROAS.
- Mobile generated the largest reported revenue and new-customer volume by device; device-level ROAS differences were comparatively small.
- Existing Customers had the highest reported audience revenue; Families had the highest CTR/conversion rate in the audience comparison.
- Monthly and quarterly results varied, while CTR and conversion rate were comparatively stable.

10 | Dashboard Evidence - Visuals 1 and 2

Figure 1. Marketing Overview Dashboard

Key finding: Google Ads generated the highest total revenue, while Email showed stronger marketing efficiency.

Interpretation: Overall performance varies by channel, so revenue and efficiency should be considered together.

Date/filter context: January 2024–December 2025; filters available for date and channel.

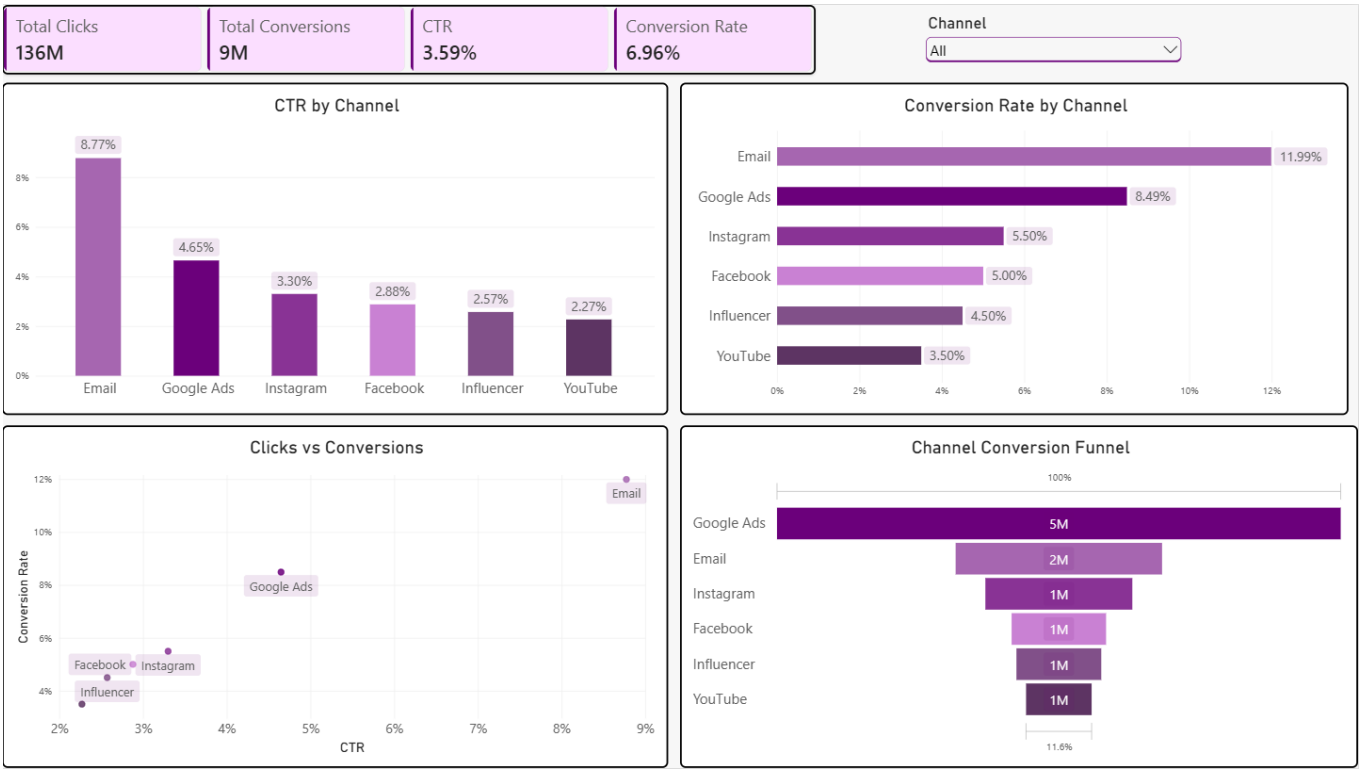


Figure 2. Channel Performance Analysis

Key finding: Email recorded the highest CTR and conversion rate, while Google Ads generated the highest total revenue.

Interpretation: Email performed strongly on engagement and conversion rates, whereas Google Ads contributed more revenue volume.

Date/filter context: January 2024–December 2025; channel performance can be compared using the available report filters.



11 | Dashboard Evidence - Visuals 3 and 4

Figure 3. Campaign Performance Analysis

Key finding: Brand Awareness generated the highest revenue among campaign types, while Retention recorded the highest ROAS.

Interpretation: The campaign type generating the most revenue is not necessarily the one delivering the highest revenue per unit of spend.

Date/filter context: January 2024–December 2025; campaign performance can be explored using the available report filters.

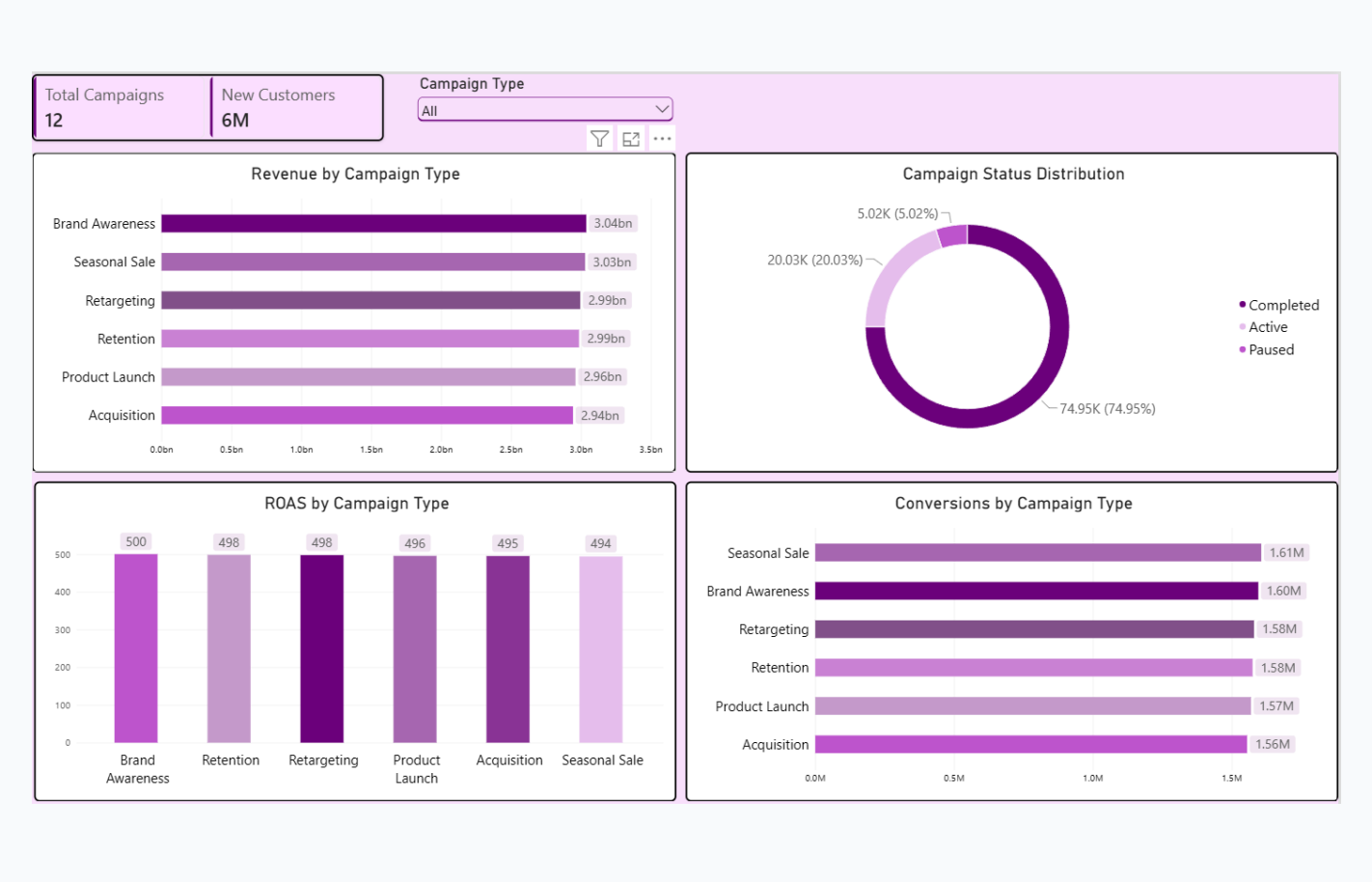


Figure 4. Regional and Audience Analysis

Key finding: West generated the highest regional revenue, while North recorded the highest regional ROAS. Existing Customers contributed the highest audience revenue.

Interpretation: Regional and audience comparisons reveal differences in revenue and efficiency that can guide further investigation.

Date/filter context: January 2024–December 2025; regional and audience comparisons use the available report filters.



12 | Recommendations & Action Plan

Priority 1 – Check Data Accuracy

The revenue and ROAS figures are unusually high. Check the data, calculations and units before making major marketing decisions.

Success measure: The totals match across Excel, PostgreSQL and Power BI, and all KPI definitions are clear.

Priority 2 – Improve Email Marketing

Email performed well in the analysis. Study which audiences and campaigns worked best, and test ways to improve results further.

Success measure: Higher conversion rates, more new customers and lower customer acquisition costs (CAC).

Priority 3 – Investigate YouTube Performance

YouTube showed weaker click and conversion performance than other channels. Review its ads, audience and targeting to find areas for improvement.

Success measure: Better click-through rates (CTR), more conversions and lower costs.

Priority 4 – Monitor Google Ads

Google Ads generated the highest revenue. Continue monitoring its spending and results to ensure the money invested is being used effectively.

Success measure: Strong revenue, reasonable acquisition costs and consistent ROAS.

Priority 5 – Test Before Changing Budgets

Before increasing or reducing marketing spend, test small changes and compare the results. Use the findings to make better budget decisions.

Success measure: Improved campaign performance and, where data is available, higher profit after costs.

13 | Limitations, Risks & Next Steps

Limitations

- The original source of the NovaMart dataset has not been confirmed.
- The data may be fictional or created for practice, so the results should not be treated as real company results.
- The exact meaning of Conversions, Revenue and New Customers has not been confirmed.
- Other factors, such as audience, season and ad quality, may also affect campaign performance.
- ROAS shows revenue earned per unit of ad spend, not actual profit.

Next Steps

1. Confirm the dataset source and usage rights.
2. Finish checking the data and make sure totals match across Excel, PostgreSQL and Power BI.
3. Add the final dashboard screenshots with their date ranges and filters.
4. Test the recommendations and check the results.
5. Organise the SQL files, Power BI report and project documentation in the portfolio repository.

14 | Conclusion & Project Handover

Final deliverable checklist

- ☑ Final Power BI dashboard saved and pages named consistently.
- ☑ Four chart screenshots added to pages 11-14 with captions.
- ☑ SQL queries saved and named by business question.
- ☑ DAX measures use consistent definitions and formats.
- ☑ Source link and attribution verified.
- ☑ Data-quality checks and caveats included.
- ☑ Recommendations tied to measurable KPIs.
- ☑ Project start and submission dates filled in.

Project details

Project prepared by: _____Darshana Narjari_____

Project start date: _____09-10-2026_____

Project submission date: _____10-10-2026_____

Portfolio / GitHub URL: _____https://github.com/darshana-analytics29/novamart-marketing-analytics

Power BI report filename: _____Marketing Analystics Dashboard____

SQL script filename: _____Marketing Analytics_____

Final takeaway: Good analytics explains what happened, checks whether the result is trustworthy, and identifies what the business should test next.